

# SHRIJAL GOSWAMI

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## SUMMARY

AI/ML Engineer specializing in LLM applications, Retrieval-Augmented Generation (RAG), and generative AI. Build and deploy production AI systems with prompt engineering, vector databases, and async FastAPI backends in Python, taking models from notebook to live REST endpoints.

## TECHNICAL SKILLS

**Languages:** Python, C++, SQL, TypeScript

**AI/ML:** LLMs, Generative AI, Agentic AI, RAG (hybrid BM25 + dense retrieval, reranking), Embeddings, Prompt Engineering, Function Calling / Tool Use, LangChain, Sentence Transformers, Hugging Face, NLP, XGBoost, PyTorch, Scikit-learn, Stacked Ensembles

**Backend/Systems:** FastAPI (async), REST API, Pydantic, Docker, Uvicorn, Vertex AI, Groq, Streamlit

**Data/Tools:** Vector Databases, ChromaDB, PyMuPDF, Pandas, NumPy, Data Pipelines, Git, GitHub

## EXPERIENCE

### AI Strategist

2026 – Present

*AKademy38 (Australia-based EdTech, Remote)*

- Architected an AI-powered Booklet Generation Engine that replaced a manual, hand-run content workflow with a structured, deterministic LLM generation pipeline serving production educational content.
- Engineered staged document generation transforming structured curriculum data into print-ready worksheets, assessments, and answer keys via automated PDF and HTML rendering.
- Reduced token usage per booklet ~10x (~40% to ~4% of prior budget) through token optimization, versioned prompt templating, staged generation, and shared-context reuse.
- Implemented automated content validation and formatting-consistency checks to keep generated output curriculum-aligned at scale.

### AI/ML Intern

2025 – 2026

*Pinnacle Labs (Kolkata, Remote)*

- Shipped the Resume Intelligence Platform as a publicly deployed live application delivering ATS scoring, skill-gap detection, and candidate ranking from NLP and semantic search.
- Designed hybrid retrieval with Sentence Transformers and BM25 for context-aware resume-job matching, owning the system end to end from data ingestion through deployment.

## PROJECTS

**Explainable RAG QA System** | [GitHub](#) | *Python, FastAPI, ChromaDB, LangChain, BM25, Groq*

- Built a source-grounded hybrid retrieval pipeline fusing BM25 sparse and dense vector search (optimal fusion weight  $\alpha=0.65$ ) for cited, hallucination-resistant question answering.
- Optimized retrieval via semantic chunking and reranking, reaching MRR 0.84 (LLM evaluation) and ~45 ms latency with incremental indexing; exposed REST APIs over FastAPI.

**Resume Parser & Job Matcher** | [GitHub](#) | [Live Demo](#) | *Python, FastAPI, NLP, Sentence Transformers, PyMuPDF*

- Engineered a multi-stage NLP pipeline (extraction → NER → taxonomy matching → parsing → scoring) converting unstructured PDF resumes into structured recruiter-ready profiles.
- Implemented semantic job matching with Sentence Transformer embeddings for automated resume-to-job scoring; deployed as a publicly accessible live application.

**Heart Disease Prediction System** | [GitHub](#) | *Python, Scikit-learn, XGBoost, FastAPI, Docker*

- Designed a stacking ensemble (Random Forest, XGBoost, logistic-regression meta-learner) reaching 89.13% CV accuracy and 0.94 ROC-AUC on the UCI Heart Disease dataset, 2.4 points over the best single model.
- Applied entropy and mutual-information feature selection with 5-fold stratified cross-validation; containerized with Docker and served real-time inference via FastAPI REST endpoints.

## EDUCATION

**Vellore Institute of Technology, Bhopal**

Sep 2024 – Oct 2028

*B.Tech in Computer Science Engineering (AI & ML)*

CGPA: 8.4

## CERTIFICATIONS & ACHIEVEMENTS

- DataForge Hackathon Finalist – IITR E-Summit 2026; GirlScript Summer of Code (GSSoC '26) Open-Source Contributor.
- Applied Machine Learning in Python – University of Michigan; Google Cloud Certifications in Gemini, Multimodal RAG, and Generative AI.